

Lab 3: Component Modelling & Architectural Pattern Selection

Project: Airport Lost Luggage Claim & Tracking Portal

Written Justification

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1. Architecture Choice

We chose Layered Architecture for the Airport Lost Luggage Claim & Tracking Portal.

2. Two Reasons

The requirements split naturally into three parts. The passenger and agent screens are the presentation layer. Claim handling, bag tag validation, cross-referencing and the 21-day compensation rule are the business layer. Claims and scan logs are stored in the data layer. Each rule sits in one place, so changing the compensation policy only affects the Compensation Service.

The payment gateway is an outside system, so only the Payment Service talks to it (IPaymentGateway). If the airline changes its payout provider, the UI and claim logic stay the same. Microservices would be too much for a system this size, and plain Client-Server would not keep business logic separate from data access.

3. Security Advantage

Only the data layer touches the database, so AES-256 encryption of passenger PII (NFR-002) is enforced in one place. The UI cannot read data directly or skip the business rules. Only the Payment Service has access to the gateway, which limits who can reach payment details.

4. Performance Benefit

The tracking timeline goes from the UI through ITrackingService straight to the data layer, without passing through claim or compensation logic. This lets scan-log queries use indexes and caching to support meeting the 200 ms target in NFR-001 during peak load.